

ANDREA RAGALZI

PYTHON BACKEND ENGINEER

Python | FastAPI | RAG

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PROFESSIONAL SUMMARY

Python Backend Engineer with around five years of experience building APIs, automation systems and production software for the insurance and industrial sectors. Specialised in Python and FastAPI, with hands-on experience in automated testing, systems integration and the full application lifecycle. Since 2025, I have continued developing my backend and applied AI skills through Document Intelligent Hub, an independent full-stack RAG project featuring advanced retrieval, conversational context management and user-scoped data isolation. Seeking a Python Backend Engineer role within an international remote or hybrid team.

INDEPENDENT SOFTWARE PROJECT

DOCUMENT INTELLIGENT HUB – Full-Stack RAG Application

2025 – Present

Independent portfolio project – [GitHub repository](#)

- Designed and developed an end-to-end RAG application for uploading, indexing and querying PDF documents, producing answers grounded in retrieved sources.
- Implemented a document-processing pipeline including text extraction, structural and fixed-size chunking, embeddings, metadata management and vector search with ChromaDB.
- Built an advanced retrieval pipeline featuring query expansion, multi-query retrieval, result deduplication and hybrid reranking.
- Implemented conversational context management through message history, query reformulation and summarisation of long conversations.
- Developed the backend with Python, FastAPI and LangChain and the frontend with Next.js and TypeScript; integrated Firebase authentication and user-scoped data isolation through metadata filtering.

PROFESSIONAL EXPERIENCE

BACKEND DEVELOPER

Oct 2023 – Feb 2025

HRM – Remote

- Developed backend services for insurance request automation using Python, AWS Lambda and hexagonal architecture.
- Maintained and optimised a production Spring Boot application within the insurance domain.
- Wrote and executed unit tests with Pytest and JUnit to prevent regressions and improve code quality.
- Collaborated remotely with international and distributed development teams to maintain and enhance production applications.

SOFTWARE DEVELOPER

Sep 2019 – Nov 2022

Gefit S.p.A. – Alessandria, Italy

- Developed Python and Flask applications for industrial computers and Raspberry Pi devices.
- Built a computer vision system using Python and YOLOv5 to automatically recognise industrial drum models.
- Designed and maintained relational databases, deployed applications to QNAP NAS infrastructure using LXC/LXD and managed the application lifecycle.
- Installed, configured and supported software at customer sites in Italy and the United States, collaborating with engineers and PLC and robotics developers.

TECHNICAL SKILLS

Backend: Python, FastAPI, Flask, REST APIs, Spring Boot, hexagonal architecture, layered architectures

RAG and applied AI: LangChain, ChromaDB, embeddings, vector search, query expansion, reranking, LLM applications, YOLOv5

Frontend: Next.js, React, TypeScript, JavaScript, HTML, CSS, Tailwind CSS

Cloud and tools: AWS Lambda, Docker, Git, Linux, Bash

Testing, databases and services: Pytest, JUnit, SQL, relational databases, MongoDB, Firebase

AI-assisted engineering workflow: ChatGPT and Codex for code exploration, testing, documentation and review, with manual validation and ownership of technical decisions

EDUCATION

Full Stack Web Developer Bootcamp

Jan 2023 – Jul 2023

Epicode Italia – Remote

University Coursework in Computer Science

2015 – 2021

Università del Piemonte Orientale – Degree programme not completed

Diploma in Electronics and Electrical Engineering

2015

IIS Ciampini-Boccardo – Novi Ligure, Italy

LANGUAGES

Italian: Native | **English:** B2 – Professional working proficiency | **Spanish:** B2 – Conversational, used daily in Málaga